Transkriptomanalyse

11. November 2026, Sommersemester 2025

Aufgabe 2

a)

Herleitung des M-Schritts für univariate Gaussche Mischmodelle:

$$\begin{aligned}
\text{M-Schritt: } Q(\theta, \theta^{(t)}, x) =& \underset{\theta}{\operatorname{argmax}} \sum_{n,k} \gamma_{n,k}^{(t)} \ln p(x_n, k|\theta) \\
\Leftrightarrow& \underset{\theta}{\operatorname{argmax}} \sum_{n,k} \gamma_{n,k}^{(t)} \ln \left(p(k|\theta) p(x_n|k, \theta)\right) \\
\Leftrightarrow& \underset{\theta}{\operatorname{argmax}} \sum_{n,k} \gamma_{n,k}^{(t)} \ln p(k|\theta) + \ln p(x_n|k, \theta) \\
\Leftrightarrow& \underset{\theta}{\operatorname{argmax}} \sum_{n,k} \gamma_{n,k}^{(t)} \ln \pi_k + \ln \left(\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x_n - \mu_k)^2}{2\sigma_k^2}} \right) \\
\Leftrightarrow& \underset{\theta}{\operatorname{argmax}} \sum_{n,k} \gamma_{n,k}^{(t)} \ln \pi_k + \ln \frac{1}{\sqrt{2\pi\sigma^2}} - \frac{(x_n - \mu_k)^2}{2\sigma_k^2}
\end{aligned}$$

Für den Fall, dass $\underline{\pi}$ und $\underline{\sigma}$ gegeben sind und $\underline{\mu}$ zu schätzen ist gilt:

$$\begin{aligned}
&\text{M-Schritt: } \underset{\underline{\hat{\mu}}}{\operatorname{argmax}} \sum_{n,k} \gamma_{n,k}^{(t)} (\ln \pi_k + \ln \frac{1}{\sqrt{2\pi\sigma^2}} - \frac{(x_n - \hat{\mu}_k)^2}{2\sigma_k^2}) \\
&\Leftrightarrow \left(\frac{\partial Q(\theta, \theta^{(t)}, x)}{\partial \hat{\mu}_k} \forall k \right)^T = 0
\end{aligned}$$

dann gilt:

$$
\begin{aligned}
&\frac{\partial Q(\theta, \theta^{(t)}, x)}{\partial \hat{\mu}_k} = 0 \\
\Leftrightarrow &\frac{\partial \sum_{n,k} \gamma_{n,k}^{(t)} (\ln \pi_k + \ln \frac{1}{\sqrt{2\pi\sigma^2}} - \frac{(x_n - \hat{\mu}_k)^2}{2\sigma_k^2})}{\partial \hat{\mu}_k} = 0 \\
&\Leftrightarrow \sum_n \gamma_{n,k}^{(t)} \frac{\partial(-\frac{(x_n - \hat{\mu}_k)^2}{2\sigma_k^2})}{\partial \hat{\mu}_k} = 0 \\
&\Leftrightarrow \sum_n -\frac{\gamma_{n,k}^{(t)}}{2\sigma_k^2} \frac{\partial (x_n - \hat{\mu}_k)^2}{\partial \hat{\mu}_k} = 0 \\
&\Leftrightarrow \sum_n \frac{\gamma_{n,k}^{(t)}}{2\sigma_k^2} 2(x_n - \hat{\mu}_k) = 0 \\
&\Leftrightarrow \sum_n \gamma_{n,k}^{(t)} (x_n - \hat{\mu}_k) = 0 \\
&\Leftrightarrow \sum_n \gamma_{n,k}^{(t)} x_n = \sum_n \gamma_{n,k}^{(t)} \hat{\mu}_k \\
&\Leftrightarrow \hat{\mu}_k = \frac{\sum_n \gamma_{n,k}^{(t)} x_n}{\sum_n \gamma_{n,k}^{(t)}}
\end{aligned}
$$

Es gilt also $\hat{\mu}_k = \frac{\sum_n \gamma_{n,k}^{(t)} x_n}{\sum_n \gamma_{n,k}^{(t)}} \quad \forall k$